Thompson's group F is nonamenable

Rewritten for readability from the OpenAI manuscript of the same title

Abstract

We prove that Thompson's group F is nonamenable. This confirms Geoghegan's conjecture and resolves the amenability problem for F.

1 The result and the plan of proof

1.1 Statement

Thompson's group F consists of the increasing piecewise linear homeomorphisms of $[0,1]$ with finitely many pieces, dyadic rational breakpoints, and slopes in $2^{\mathbb{Z}}$. Products are compositions, $hg = h \circ g$, and F is regarded as a discrete group.

A discrete group Λ is *amenable* if there is a linear functional $\mu : \ell^\infty(\Lambda;\mathbb{R}) \to \mathbb{R}$ with $\mu(1) = 1$, $\mu(\varphi) \geq 0$ whenever $\varphi \geq 0$, and $\mu(g \mapsto \varphi(hg)) = \mu(\varphi)$ for every $h \in \Lambda$ [19, 6]. We use only the forward implication of the Følner criterion [7]: if Λ is amenable, then for every finite $S \subset \Lambda$ and every $\varepsilon > 0$ there is a nonempty finite $A \subset \Lambda$ with

$$\frac{|hA \triangle A|}{|A|} < \varepsilon \qquad (h \in S). \tag{1.1}$$

(The left-translation formulation in [14, Theorem 1.2] bounds the sum of these ratios and hence implies each strict individual bound by taking a smaller tolerance.)

Theorem 1.1. *Thompson's group F is not amenable.*

The only other outside input is the following.

Lemma 1.2 (Benyamini–Sternfeld)**.** *There are a real Hilbert space H, a Lipschitz map $f : B \to B$ on its closed unit ball, and a constant $\delta > 0$ such that $\|f(x) - x\| \geq \delta$ for all $x \in B$.*

Benyamini and Sternfeld prove this for every infinite-dimensional normed space [2, Theorem, part (3), p. 439]. Appendix A gives a self-contained construction on $L^2([0,1];\mathbb{R}^2)$ with $\delta = 1/2$. In finite dimensions the lemma is false (Brouwer), so the infinite-dimensional space is essential.

Context. Thompson introduced F in 1965; an early published construction is in [11], with the identification explained in [5, pp. 215–216]. Geoghegan's 1979 conjecture that F is nonamenable is recorded in [5, p. 227]. The problem is delicate because F has no nonabelian free subgroup [3, Theorem 3.1] yet is not elementary amenable [5, Theorem 4.10]; it also has a classifying space with finitely many cells in each dimension [4]. Earlier constraints on amenability come from sizes of Følner sets [15, Theorem 1.1], a convex Ramsey characterization for binary trees [14, Theorem 3.1], densities and diagram classes [8, 9], and the reduced C^*-algebra of Thompson's group T [10, Theorem 5.5]. Earlier claimed resolutions: nonamenability [1]; amenability [18], with errors identified in [12]; and a withdrawn amenability claim [13] (a different work from [14]). None of this is used below.

1.2 Proof at a glance

Fix f, a Lipschitz constant L and δ from Lemma 1.2, and an integer $D > 4L^2/\delta^2$.

Objects. A basic interval is $[k2^{-r}, (k+1)2^{-r}]$; a basic partition is a finite partition of $[0,1]$ into basic intervals. Fix D basic intervals $I_1 < \cdots < I_D$ ("parents") with gaps between them and endpoints in $(0,1)$, and inside each parent I_i the affine copies $I_i \cdot I_j$ of all parents ("children"). Every basic partition T gets a *color* $p(T) \in B$ by recursion on the number of cells: if every parent is a union of cells of T, then $p(T)$ is f applied to the mean of the colors of the D partitions obtained by restricting T to a parent and rescaling to $[0,1]$; otherwise $p(T) = 0$. For $g \in F$ and n large, the image $gT^{(n)}$ of the uniform partition of mesh 2^{-n} is basic, and $X_I(g) \in B$ denotes the color of its rescaled restriction to I.

Key idea. These vectors have two exact properties:

(a) *Recursion:* $X_{I_i}(g) = f(z_i(g))$, where $z_i(g) = \frac{1}{D}\sum_j X_{I_i \cdot I_j}(g)$.

(b) *Transport:* for intervals $I < J$ of the family some $h_{I,J} \in F$ maps I, J affinely onto I_1, I_2, and then $X_I(g) = X_{I_1}(h_{I,J}g)$, $X_J(g) = X_{I_2}(h_{I,J}g)$.

Let S be the finite set of all $h_{I,J}$; it is fixed before any finite set $A \subset F$ is considered. Put $\eta = \max_{h \in S} |hA \triangle A|/|A|$. By (b), the A-average of $\langle X_I, X_J \rangle$ is within η of one number α for *all* separated pairs. Expanding the A-average of $\|z_i - m\|^2$, where $m = \frac{1}{D}\sum_k X_{I_k}$, the α's cancel and the non-separated pairs form a fraction $1/D$ of the terms, so it is at most $4/D + 4\eta$. By (a), m is the mean of the $f(z_i)$, so $\|m - f(m)\|^2 \le \frac{L^2}{D}\sum_i \|z_i - m\|^2$. Averaging, $\delta^2 \le L^2(4/D + 4\eta)$: a positive lower bound for η independent of A, contradicting (1.1).

Dependencies.

- §2.1: restriction composes (2.1); fine images are basic (Lemma 2.1); pair transport (Lemma 2.2); covariance (2.2).
- §2.2: the coloring, using f.
- §2.3: Proposition 2.3 (the bound on η). Property (b) uses Lemmas 2.1, 2.2 and (2.2); (a) uses (2.1); the contradiction uses Lemma 1.2. Theorem 1.1 follows from Proposition 2.3 and (1.1).
- Appendix A proves Lemma 1.2; §3 (consequences) is not used in the proof.

Renamed symbols. The source reuses several letters. Here the generic group is Λ and its mean μ; in §3 the generating set is Σ, the Hilbert space $\mathcal{K}$, the conjugating operator V, the connection kernel $\mathcal{T}_p$; in the appendix $\mathrm{Lip}(\gamma')$ is K_γ and the rotated field Jv is written $v^\perp$.

2 Recursive colors and finite averages

2.1 Dyadic partitions and affine transport

We use the standard dyadic-partition model of F [5, Section 2, Lemma 2.2].

A *basic interval* is $[k2^{-r}, (k+1)2^{-r}]$ with integers $r \ge 0$, $0 \le k < 2^r$. A *basic partition* is a finite set of basic intervals (its *cells*) covering $[0,1]$ with pairwise disjoint interiors; its *mesh* is the largest cell length. A partition T *respects* an interval I if I is a union of cells of T. We use the standard fact that two basic intervals whose interiors meet are nested (dyadic intervals form a binary tree under inclusion).

For a basic interval $I = [a2^{-r}, (a+1)2^{-r}]$ let $s_I(x) = 2^{-r}(x + a)$, the increasing affine bijection $[0,1] \to I$, and write $I \cdot J = s_I(J)$. If T respects I, its *normalized restriction* is

$$T_I = \{s_I^{-1}(C) : C \in T,\ C \subseteq I\}.$$

Basicness. s_I maps $[b2^{-t},(b+1)2^{-t}]$ onto $[(a2^t+b)2^{-(r+t)},(a2^t+b+1)2^{-(r+t)}]$, so $I\cdot J$ is basic. A basic $C\subseteq I$ is $[k2^{-s},(k+1)2^{-s}]$ with $s\geq r$, and $s_I^{-1}(C)$ has left endpoint $(k-a2^{s-r})2^{-(s-r)}$ and length $2^{-(s-r)}$, so it is basic. Hence T_I is a basic partition.

Restriction composes. Both $s_{I\cdot J}$ and $s_I\circ s_J$ are the increasing affine bijection $[0,1]\to I\cdot J$, hence equal. Suppose T respects I. Applying s_I shows that J is a union of cells $s_I^{-1}(C)$ of T_I exactly when $I\cdot J$ is a union of cells C of T. Thus T_I respects J if and only if T respects $I\cdot J$, and then

$$(T_I)_J=\{s_J^{-1}s_I^{-1}(C):C\in T,\ C\subseteq I\cdot J\}=T_{I\cdot J}. \tag{2.1}$$

Let $T^{(n)}$ be the uniform partition into intervals of length 2^{-n}, and for $g\in F$ let $gT^{(n)}$ be the set of images of its cells.

Lemma 2.1. *For each $g\in F$, the partitions $gT^{(n)}$ are basic for all sufficiently large n, and their meshes tend to zero. In particular, given a finite set $K\subset F$ and a finite family $\mathcal{J}$ of basic intervals, there is an n_0 such that $gT^{(n)}$ is basic and respects every member of $\mathcal{J}$ whenever $g\in K$ and $n\geq n_0$.*

This lemma lets one level n serve all finitely many group elements in Proposition 2.3.

Proof. Step 1. Every linear piece of g is $x\mapsto 2^qx+b$ with $q\in\mathbb{Z}$ and b dyadic: starting from $g(0)=0$, the image of each breakpoint is a sum of increments (power of two times dyadic length), hence dyadic.

Step 2. Write $b=c2^{-N}$ with $c\in\mathbb{Z}$, $N\geq 0$. Let n be so large that all breakpoints are multiples of 2^{-n} and $n\geq q+N$ on every piece. Each cell $[k2^{-n},(k+1)2^{-n}]$ then lies in one piece and its image is $[(k+c2^{n-q-N})2^{q-n},(k+c2^{n-q-N}+1)2^{q-n}]\subseteq[0,1]$, a basic interval of length 2^{q-n}. So $gT^{(n)}$ is basic with mesh at most $2^{q_{\max}-n}\to 0$.

Step 3. A basic partition with mesh smaller than the length of a basic interval I respects I: every cell whose interior meets that of I is nested with I and, being shorter, lies in I; these cells cover I.

Step 4. Apply Step 3 to each member of $\mathcal{J}$ and take the largest of the finitely many thresholds over $g\in K$. □

Write $I<J$ when $\max I<\min J$ (a positive gap is required). An interval is *internal* if both endpoints lie in $(0,1)$.

Lemma 2.2. *If $I<J$ and $I'<J'$ are pairs of internal basic intervals, there is an $h\in F$ whose restrictions carry I affinely onto I' and J affinely onto J'.*

This lemma supplies the finite set S in §2.3.

Proof. Step 1. The complement of $I\cup J$ consists of three gaps $[0,\min I]$, $[\max I,\min J]$, $[\max J,1]$, each of positive length (internal, $I<J$) with dyadic endpoints; likewise for I',J'.

Step 2. Cut each of the six gaps into basic intervals, e.g. the cells of a sufficiently fine uniform dyadic grid.

Step 3. Bisecting a basic interval gives two basic intervals and raises the count by one. Bisecting on the side with fewer cells, make the k-th source gap and the k-th target gap have equally many cells ($k=1,2,3$).

Step 4. Now $[0,1]$ is cut on both sides into the same number of basic intervals, with I,J in the same positions as I',J'. Let h map each source interval affinely and increasingly onto the corresponding target interval. Adjacent pieces agree at shared endpoints, so h is an increasing homeomorphism of $[0,1]$; its breakpoints are dyadic and each slope is a ratio of two basic lengths, a power of two. Hence $h\in F$, and it is affine from I onto I' and from J onto J'. □

Covariance. Let $h \in F$ satisfy $h(I) = I'$ with $h|_I$ affine. Then $h \circ s_I = s_{I'}$ (both are the increasing affine bijection $[0,1] \to I'$). If $gT^{(n)}$ and $(hg)T^{(n)}$ are basic and respect I and I' respectively, then

$$\big((hg)T^{(n)}\big)_{I'} = (gT^{(n)})_I \tag{2.2}$$

as sets of intervals. Indeed, the cells of $(hg)T^{(n)}$ are the images $h(C)$ of the cells C of $gT^{(n)}$; since h is a bijection with $h(I) = I'$, $h(C) \subseteq I'$ exactly when $C \subseteq I$; and $s_{I'}^{-1}(h(C)) = s_I^{-1}(C)$ for such C.

2.2 The recursive coloring

Fix H, B, f as in Lemma 1.2, and $L > 0$, $\delta > 0$ with

$$\|f(x) - f(y)\| \le L\|x - y\|, \qquad \|f(x) - x\| \ge \delta \quad (x, y \in B). \tag{2.3}$$

Choose an integer $D \ge 2$ with

$$D > \frac{4L^2}{\delta^2}, \tag{2.4}$$

and internal basic intervals $I_1 < \cdots < I_D$; for instance, with $2^r > 2D$, $I_j = [(2j-1)2^{-r}, 2j2^{-r}]$.

Define $p(T) \in B$ for every basic partition T by induction on its number of cells:

$$p(T) = \begin{cases} f\Big(\dfrac{1}{D}\displaystyle\sum_{j=1}^{D} p(T_{I_j})\Big), & \text{if } T \text{ respects every } I_j, \\ 0, & \text{otherwise.} \end{cases} \tag{2.5}$$

This is well defined: in the first case T has a cell outside $I_j \ne [0,1]$, so T_{I_j} has strictly fewer cells and its color is already defined; the mean of points of the convex set B lies in B, the domain of f; and f maps into B.

Sanity check. For the displayed I_j: $p(T^{(0)}) = 0$ (second case); $(T^{(r)})_{I_j} = T^{(0)}$, so $p(T^{(r)}) = f(0)$; $(T^{(2r)})_{I_j} = T^{(r)}$, so $p(T^{(2r)}) = f(f(0))$.

Let

$$\mathcal{I} = \{I_i : 1 \le i \le D\} \cup \{I_i \cdot I_j : 1 \le i, j \le D\},$$

a finite family of internal basic intervals. Call (n, g) *admissible* if $gT^{(n)}$ is a basic partition respecting every member of $\mathcal{I}$. For all n, $g \in F$, $I \in \mathcal{I}$ put

$$X_I^{(n)}(g) = \begin{cases} p\big((gT^{(n)})_I\big), & \text{if } (n,g) \text{ is admissible,} \\ 0, & \text{otherwise,} \end{cases} \qquad \varphi_{I,J}^{(n)}(g) = \langle X_I^{(n)}(g), X_J^{(n)}(g)\rangle. \tag{2.6}$$

These are defined on all of F, with $X_I^{(n)}(g) \in B$ and $|\varphi_{I,J}^{(n)}| \le 1$. By Lemma 2.1 (with $\mathcal{J} = \mathcal{I}$), for every finite $K \subset F$ there is one n with (n,g) admissible for all $g \in K$.

2.3 A uniform boundary bound

For every ordered pair $I < J$ in $\mathcal{I}$ fix, by Lemma 2.2, an $h_{I,J} \in F$ carrying I, J affinely onto I_1, I_2. Let S be the finite set of these elements. It depends only on D and the I_j, not on any set A.

Finite-average comparison. For nonempty finite $A \subset F$, $h \in F$ and $\varphi : F \to [-1,1]$,

$$\Big|\frac{1}{|A|}\sum_{g\in A}\varphi(hg) - \frac{1}{|A|}\sum_{g\in A}\varphi(g)\Big| \le \frac{|hA \mathbin{\triangle} A|}{|A|}. \tag{2.7}$$

Indeed $\sum_{g\in A}\varphi(hg) = \sum_{x\in hA}\varphi(x)$; the terms with $x \in hA \cap A$ cancel against the second sum, leaving $|hA \setminus A| + |A \setminus hA|$ terms of absolute value at most one.

Proposition 2.3. *For every nonempty finite set $A \subset F$,*

$$\max_{h\in S} \frac{|hA \triangle A|}{|A|} \geq \frac{\delta^2/L^2 - 4/D}{4(1-1/D)} > 0. \tag{2.8}$$

Plan. Steps 1–2 establish properties (b) and (a) of the overview; Step 3 is a pointwise inequality from (2.3); Step 4 bounds an averaged variance; Step 5 combines.

Proof. Step 0 (setup). Fix A and put

$$\eta = \max_{h\in S} \frac{|hA \triangle A|}{|A|}. \tag{2.9}$$

The set $A \cup SA$, $SA = \bigcup_{h\in S} hA$, is finite, so there is one n with (n, g) admissible for all $g \in A \cup SA$. This n depends on A and is fixed for the rest of the proof; *all* colors and correlations below are taken at this level and the superscript is omitted. Write

$$\mathbb{E}_A\psi = \frac{1}{|A|}\sum_{g\in A} \psi(g), \qquad \alpha = \mathbb{E}_A\varphi_{I_1,I_2}.$$

The number α depends on A and n; it will cancel.

Step 1 (all separated correlations are near α). Let $I < J$ in $\mathcal{I}$, $h = h_{I,J}$, $g \in A$. Both (n, g) and (n, hg) are admissible, so $gT^{(n)}$ is basic and respects I, J, and $(hg)T^{(n)}$ is basic and respects I_1, I_2. By (2.2) (for $I \to I_1$ and for $J \to I_2$) and (2.6),

$$X_{I_1}(hg) = p\big(((hg)T^{(n)})_{I_1}\big) = p\big((gT^{(n)})_I\big) = X_I(g), \qquad X_{I_2}(hg) = X_J(g).$$

Hence $\varphi_{I,J}(g) = \varphi_{I_1,I_2}(hg)$ for $g \in A$. Averaging and applying (2.7) to the globally defined φ_{I_1,I_2}, then (2.9):

$$\big|\mathbb{E}_A\langle X_I, X_J\rangle - \alpha\big| \leq \eta \qquad \text{whenever } I < J \text{ or } J < I \text{ in } \mathcal{I} \tag{2.10}$$

(the case $J < I$ by symmetry of the inner product). The separated pairs used below are: (i) I_k, I_l with $k \neq l$; (ii) $I_i \cdot I_j$, $I_i \cdot I_k$ with $j \neq k$ (s_{I_i} is increasing, so it preserves order and gaps); (iii) $I_i \cdot I_j$, I_k with $k \neq i$ (because $I_i \cdot I_j \subseteq I_i$). Nothing is claimed for $I_i \cdot I_j$ against I_i, nor for an interval against itself.

Step 2 (recursion). For $g \in A$ put

$$m(g) = \frac{1}{D}\sum_{k=1}^{D} X_{I_k}(g), \qquad z_i(g) = \frac{1}{D}\sum_{j=1}^{D} X_{I_i\cdot I_j}(g) \quad (1 \leq i \leq D),$$

points of B by convexity. Since $gT^{(n)}$ respects I_i and every $I_i \cdot I_j$, (2.1) shows that $(gT^{(n)})_{I_i}$ respects every I_j, with $((gT^{(n)})_{I_i})_{I_j} = (gT^{(n)})_{I_i\cdot I_j}$. So the first case of (2.5) applies to $T = (gT^{(n)})_{I_i}$:

$$X_{I_i}(g) = f(z_i(g)) \qquad (g \in A,\ 1 \leq i \leq D). \tag{2.11}$$

Step 3 (pointwise inequality). Fix $g \in A$ and drop it from the notation. By (2.11), $m = \frac{1}{D}\sum_i f(z_i)$, so

$$\begin{aligned}\delta^2 \leq \|m - f(m)\|^2 &= \Big\|\frac{1}{D}\sum_{i=1}^{D}(f(z_i) - f(m))\Big\|^2 \\ &\leq \frac{1}{D}\sum_{i=1}^{D}\|f(z_i) - f(m)\|^2 \leq \frac{L^2}{D}\sum_{i=1}^{D}\|z_i - m\|^2,\end{aligned} \tag{2.12}$$

using, in order, the displacement bound in (2.3), convexity of $\|\cdot\|^2$, and the Lipschitz bound in (2.3).

Step 4 (variance). Expand $\|m\|^2 = D^{-2}\sum_{k,l}\langle X_{I_k}, X_{I_l}\rangle$. The $D(D-1)$ terms with $k \neq l$ are of type (i), each with average at most $\alpha+\eta$ by (2.10); the D diagonal terms are at most one. The same holds for $\|z_i\|^2$ with type (ii). So

$$\mathbb{E}_A\|m\|^2 \le \Big(1-\frac{1}{D}\Big)(\alpha+\eta)+\frac{1}{D}, \tag{2.13}$$

$$\mathbb{E}_A\|z_i\|^2 \le \Big(1-\frac{1}{D}\Big)(\alpha+\eta)+\frac{1}{D}. \tag{2.14}$$

In $\langle z_i, m\rangle = D^{-2}\sum_{j,k}\langle X_{I_i\cdot I_j}, X_{I_k}\rangle$ the $D(D-1)$ terms with $k \neq i$ are of type (iii), each with average at least $\alpha-\eta$; the D terms with $k=i$ (a child against its own parent) are at least -1. So

$$\mathbb{E}_A\langle z_i, m\rangle \ge \Big(1-\frac{1}{D}\Big)(\alpha-\eta)-\frac{1}{D}. \tag{2.15}$$

These bounds hold for α, η of any sign or size. From $\|z_i-m\|^2 = \|z_i\|^2+\|m\|^2-2\langle z_i, m\rangle$,

$$\mathbb{E}_A\|z_i-m\|^2 \le 2\Big(1-\frac{1}{D}\Big)(\alpha+\eta)+\frac{2}{D}-2\Big(1-\frac{1}{D}\Big)(\alpha-\eta)+\frac{2}{D} = \frac{4}{D}+4\Big(1-\frac{1}{D}\Big)\eta. \tag{2.16}$$

Step 5 (conclusion). Average (2.12) over $g \in A$ and insert (2.16):

$$\delta^2 \le \frac{L^2}{D}\sum_{i=1}^{D}\mathbb{E}_A\|z_i-m\|^2 \le L^2\Big(\frac{4}{D}+4\Big(1-\frac{1}{D}\Big)\eta\Big).$$

Since $D \ge 2$, solving for η gives (2.8); the right-hand side is positive by (2.4) and involves neither A nor n. □

Proof of Theorem 1.1. Let $c_0 > 0$ be the right-hand side of (2.8). If F were amenable, (1.1) with the finite set S and $\varepsilon = c_0$ would give a nonempty finite A with $\max_{h\in S}|hA\triangle A|/|A| < c_0$, contradicting Proposition 2.3. □

Remark 2.4 (What is used about F)**.** The proof of Proposition 2.3 uses F only through Step 1 (transport identities on A) and Step 2 (recursion on A), at one level n. Sanity check: if every $h_{I,J}$ were the identity, Step 1 would force all X_I to coincide, and Step 2 would give a fixed point of f, which is impossible; this matches (2.8), which would then read $0 \ge c_0$. The content specific to F is that both properties hold with the transports of Lemma 2.2, which change the size and position of intervals, while the recursion stays well founded.

3 Consequences

Two corollaries are deduced in the source from companion results; neither is used above.

Corollary 3.1. *For every $\varepsilon > 0$, there are a separable complex Hilbert space $\mathcal{K}$ and a representation $\pi : F \to \mathrm{GL}(\mathcal{K})$ such that $\sup_{g\in F}\|\pi(g)\| \le 1+\varepsilon$, but no bounded invertible operator V on $\mathcal{K}$ makes every $V\pi(g)V^{-1}$ unitary.*

Proof. F is countable (each element is given by finitely many dyadic breakpoints and slopes) and nonamenable by Theorem 1.1; the source concludes by the companion unitarizability theorem [17, Theorem 1.1].

Unverified step. The source does not reproduce the statement of the companion theorem, so I could not check that countability and nonamenability are its only hypotheses or that it yields the bound $1+\varepsilon$. A theorem of that generality would settle Dixmier's unitarizability problem, which I know only as open.

□

For a finite symmetric generating set $\Sigma \subset F \setminus \{\mathrm{id}\}$, let $\mathcal{G}_\Sigma$ be the simple undirected Cayley graph with edges $\{x, xs\}$, $s \in \Sigma$. In Bernoulli bond percolation each edge is independently open with probability p. Let p_c be the threshold for existence of an infinite open cluster and p_u the threshold for almost-sure uniqueness of the infinite open cluster. The connection kernel $\mathcal{T}_p(x,y) = \mathbb{P}_p(x \leftrightarrow y)$ acts on counting-measure $\ell^2(F)$, initially on finitely supported functions by summation over y; put $p_{2\to 2} = \sup\{p \in [0,1] : \mathcal{T}_p \text{ is bounded on } \ell^2(F)\}$.

Corollary 3.2. *For every such generating set Σ,*

$$\|\mathcal{T}_{p_c}\|_{2\to 2} < \infty, \qquad p_c < p_{2\to 2} \le p_u.$$

For each fixed $p \in (p_c, p_u)$, there are almost surely infinitely many infinite open clusters. Moreover, there are deterministic $p_c < p_1 < p_2 < 1$, depending on Σ, such that in the coupling with independent uniform $[0,1]$ edge labels U_e, opening e when $U_e \le p$, there are almost surely infinitely many infinite open clusters simultaneously for every $p \in [p_1, p_2]$.

Proof. The source applies the companion percolation result [16, Corollary 1.2] to F, nonamenable by Theorem 1.1.

Unverified step. The statement of the companion corollary is not reproduced in the source, so none of the conclusions can be checked from this document. The general inequality $p_c < p_u$ for nonamenable Cayley graphs is, to my knowledge, an open conjecture.

□

A An explicit map with positive displacement

This appendix proves the Hilbert-space case of Lemma 1.2 [2] directly. It is independent of §2; here r denotes a distance.

Proposition A.1. *Let $H = L^2([0,1];\mathbb{R}^2)$ be a real Hilbert space and let $B = \{x \in H : \|x\| \le 1\}$. There is a globally Lipschitz map $f : B \to B$ such that*

$$\|f(x)\| = 1, \qquad \|f(x) - x\| \ge \frac{1}{2} \quad (x \in B).$$

Plan. We build a Lipschitz $G : B \to H$ with $\|G(x)\| \ge \rho/4$ and $G(x) = x$ for $\|x\| \ge 1/2$, and set $f = -G/\|G\|$. G is the identity outside a thin tube around a curve γ through the origin. The curve enters along a straight ray and then spirals forever inside a small ball without approaching itself (Lemma A.2); inside the tube, G moves points along the curve direction so that 0 is avoided (Lemma A.3 supplies a needed rotation).

Define

$$a(t) = \begin{cases} t/8, & t \le 2, \\ 5/16 - (3-t)^2/16, & 2 < t < 3, \\ 5/16, & t \ge 3, \end{cases} \qquad \theta(t) = \begin{cases} 0, & t \le 1, \\ (t-1)^2/2, & 1 < t < 2, \\ t - 3/2, & t \ge 2, \end{cases}$$

and, for $w \in [0,1]$,

$$v(t)(w) = (\cos(\theta(t)w), \sin(\theta(t)w)), \qquad \gamma(t) = a(t)v(t).$$

Elementary facts. The pieces match to first order at the junctions ($a(2) = 1/4$, $a'(2) = 1/8$, $a'(3) = 0$; $\theta(2) = 1/2$, $\theta'(1) = 0$, $\theta'(2) = 1$), so a, θ are C^1 with Lipschitz derivatives, $0 \le a' \le 1/8$, $0 \le \theta' \le 1$. Moreover $a \le 5/16$; $a(t) \ge 1/8$ for $t \ge 1$ and $a(t) \ge 1/4$ for $t \ge 2$; $\theta = 0$ on $(-\infty, 1]$ and θ is strictly increasing on $[1, \infty)$. Also $\|v(t)\| = 1$, $\|\gamma(t)\| = |a(t)|$, $\gamma(0) = 0$, and for $t \le 1$: $v(t) = (1,0)$ and $\gamma(t) = (t/8)(1,0)$.

Lemma A.2. *The curve γ is injective, has a bounded Lipschitz derivative, and*

$$c := \inf_{t \in \mathbb{R}} \|\gamma'(t)\| = \frac{1}{8}.$$

Its unit tangent $u(t) = \gamma'(t)/\|\gamma'(t)\|$ is Lipschitz, $\langle u(t), v(t)\rangle \ge 0$, and $u(t) = v(t)$ for $t \le 1$. There is a radius $0 < \rho < 1/32$ such that every point in

$$U = \{x \in H : \operatorname{dist}(x, \gamma(\mathbb{R})) < \rho\}$$

has a unique nearest point $\gamma(t(x))$. Writing $e(x) = x - \gamma(t(x))$, we have $e(x) \perp u(t(x))$, and

$$|t(x) - t(y)| \le 16\|x - y\| \quad \text{if } x, y \in U \text{ and } \|x - y\| < \rho. \tag{A.1}$$

Proof. Step 1 (derivative and speed). Let $v^\perp(t)(w) = (-\sin(\theta(t)w), \cos(\theta(t)w))$. For each w,

$$\partial_t(\gamma(t)(w)) = a'(t)v(t)(w) + a(t)\theta'(t)\, w\, v^\perp(t)(w).$$

This is locally Lipschitz in t uniformly in $w \in [0,1]$, so the difference quotients converge uniformly in w, hence in H; the right side is $\gamma'(t)$. As $v \perp v^\perp$ pointwise and $\int_0^1 w^2 dw = 1/3$,

$$\|\gamma'(t)\|^2 = a'(t)^2 + \frac{a(t)^2\theta'(t)^2}{3}, \qquad \langle \gamma'(t), v(t)\rangle = a'(t).$$

For $t \le 2$, $a' = 1/8$, so $\|\gamma'\| \ge 1/8$, with equality for $t \le 1$ (where $\theta' = 0$). For $t \ge 2$, $a \ge 1/4$ and $\theta' = 1$, so $\|\gamma'\|^2 \ge 1/48 > 1/64$. Hence $c = 1/8$.

Step 2 (regularity). Pointwise $|v(t)(w) - v(s)(w)| \le |\theta(t) - \theta(s)|w$, so $\|v(t) - v(s)\| \le |t - s|$; likewise for $wv^\perp$. The coefficients a' and $a\theta'$ are bounded and Lipschitz ($a\theta'$ vanishes for $t \le 1$, and for $t \ge 1$ both factors are bounded and Lipschitz). So γ' is bounded, by M say, and Lipschitz, with constant K_γ. Since $\|\gamma'\| \ge 1/8$, u is Lipschitz, with constant K_u. Finally $\langle u, v\rangle = a'/\|\gamma'\| \ge 0$, and for $t \le 1$, $\gamma' = v/8$, so $u = v$.

Step 3 (injectivity). With $\operatorname{sinc}(q) = \sin(q)/q$, $\operatorname{sinc}(0) = 1$,

$$\langle v(s), v(t)\rangle = \int_0^1 \cos((\theta(t) - \theta(s))w)dw = \operatorname{sinc}(\theta(t) - \theta(s)).$$

Since $|\operatorname{sinc}(q)| < 1$ for $q \ne 0$, the unit vectors $v(s), v(t)$ are linearly independent whenever $\theta(s) \ne \theta(t)$ (equality case of Cauchy–Schwarz). Let $s < t$. If $t \le 1$, $\gamma(s) = (s/8)(1,0) \ne \gamma(t)$. If $t > 1$, then $\theta(s) < \theta(t)$, so $a(s)v(s) = a(t)v(t)$ would force $a(t) = 0$, contradicting $a(t) \ge 1/8$.

Step 4 (uniform separation). We claim

$$\sigma(d) := \inf_{|t-s| \ge d} \|\gamma(t) - \gamma(s)\| > 0 \qquad (d > 0). \tag{A.2}$$

Otherwise there are $s_n < t_n$ with $t_n - s_n \ge d$ and $\|\gamma(t_n) - \gamma(s_n)\| \to 0$; pass to a subsequence with $s_n \to s_*$, $t_n \to t_*$ in $[-\infty, \infty]$. Each case is contradictory:

- Both finite: $t_* - s_* \geq d$ and $\gamma(s_*) = \gamma(t_*)$, against Step 3.
- $t_* = -\infty$: eventually $s_n < t_n \leq 1$ and the distance is $(t_n - s_n)/8 \geq d/8$.
- $s_* = -\infty < t_*$: $\|\gamma(s_n)\| = |s_n|/8 \to \infty$ while $\|\gamma(t_n)\|$ stays bounded.
- s_* finite, $t_* = +\infty$: $\theta(t_n) - \theta(s_n) \to \infty$, so $\langle v(s_n), v(t_n)\rangle \to 0$ and $\|\gamma(t_n) - \gamma(s_n)\|^2 \to (5/16)^2 + a(s_*)^2 > 0$.
- $s_* = +\infty$: eventually $s_n, t_n \geq 3$, so $a = 5/16$, $\theta(t_n) - \theta(s_n) = t_n - s_n \geq d$, and

$$\|\gamma(t_n) - \gamma(s_n)\|^2 = 2(5/16)^2(1 - \operatorname{sinc}(t_n - s_n)) \geq 2(5/16)^2(1 - \sup_{q \geq d} \operatorname{sinc}(q)) > 0,$$

  since sinc is continuous, less than 1 on $[d, \infty)$, and tends to 0.

Step 5 (tangent estimate; choice of ρ). Fix $d > 0$ with $K_\gamma d \leq c/2$. Integrating $\gamma'(\tau) - \gamma'(s)$ from s to t,

$$\|\gamma(t) - \gamma(s) - (t-s)\gamma'(s)\| \leq \frac{K_\gamma}{2}|t-s|^2.$$

Pair with $u(s)$ and use $\langle \gamma'(s), u(s)\rangle = \|\gamma'(s)\| \geq c$:

$$|\langle \gamma(t) - \gamma(s), u(s)\rangle| \geq c|t-s| - \frac{K_\gamma}{2}|t-s|^2 \geq \frac{3c}{4}|t-s| \qquad (|t-s| \leq d). \tag{A.3}$$

Choose $\rho > 0$ with

$$\rho < \frac{1}{32}, \qquad 4\rho < \sigma(d), \qquad \rho K_u < \frac{c}{4}. \tag{A.4}$$

Step 6 (a nearest point exists; orthogonality). Let $x \in U$ and let $\gamma(t_n)$ be a minimizing sequence for $\operatorname{dist}(x, \gamma(\mathbb{R}))$. Late terms satisfy $\|x - \gamma(t_n)\| < 2\rho$, so they are pairwise less than $4\rho < \sigma(d)$ apart, and by (A.2) their parameters differ by less than d. So the late t_n lie in a compact interval, and a convergent subsequence attains the minimum. At a minimizing t, the derivative of $\|x - \gamma(t)\|^2$ vanishes: $(x - \gamma(t)) \perp \gamma'(t)$, i.e. $\perp u(t)$.

Step 7 (parameter estimate; uniqueness). Let $x, y \in U$, $\|x - y\| < \rho$, with minimizing parameters t, s and $x = \gamma(t) + e_t$, $y = \gamma(s) + e_s$, $\|e_t\|, \|e_s\| < \rho$. Then $\|\gamma(t) - \gamma(s)\| < 3\rho < \sigma(d)$, so $|t - s| < d$. Since $e_s \perp u(s)$ and $e_t \perp u(t)$,

$$\langle x - y, u(s)\rangle = \langle \gamma(t) - \gamma(s), u(s)\rangle + \langle e_t, u(s) - u(t)\rangle,$$

so by (A.3) and (A.4)

$$\|x - y\| \geq \Big(\frac{3c}{4} - \rho K_u\Big)|t-s| \geq \frac{c}{2}|t-s| = \frac{1}{16}|t-s|.$$

Taking $y = x$ gives uniqueness of the minimizing parameter; in general this is (A.1). □

Lemma A.3. *There is a family of orthogonal operators $R_t : H \to H$, Lipschitz in operator norm, such that $R_t u(t) = v(t)$ and R_t is the identity for $t \leq 1$.*

It is used to turn a residual $e \perp u(t)$ into a vector $R_t e \perp v(t)$.

Proof. Suppress t and put $\kappa = \langle u, v\rangle \geq 0$ (Lemma A.2). Define

$$Ah = v\langle u, h\rangle - u\langle v, h\rangle, \qquad R = \mathrm{Id} + A + \frac{A^2}{1+\kappa}.$$

A is skew-adjoint, since $\langle Ah, k\rangle = \langle v, k\rangle\langle u, h\rangle - \langle u, k\rangle\langle v, h\rangle = -\langle h, Ak\rangle$. If u, v are independent, compute on their span: $Au = v - \kappa u$, $Av = \kappa v - u$, hence $A^2 u = -(1 - \kappa^2)u$, $A^2 v = -(1 - \kappa^2)v$,

so $R = \kappa \mathrm{Id} + A$ there. Then $Ru = v$, and $(\kappa - A)(\kappa + A) = \kappa^2 - A^2 = \mathrm{Id}$ on the span, so R is orthogonal there; on the orthogonal complement $A = 0$ and $R = \mathrm{Id}$. If u, v are dependent then $u = v$ (as $\kappa \geq 0$), $A = 0$, $R = \mathrm{Id}$; this covers $t \leq 1$. Finally A_t and $\kappa(t)$ are Lipschitz because u, v are, $\|A_t\| \leq 2$, and $1 + \kappa \geq 1$, so $t \mapsto R_t$ is Lipschitz in operator norm, with constant K_R say. □

Proof of Proposition A.1. Fix ρ, U, $t(\cdot)$, $e(\cdot)$ from Lemma A.2 and R_t from Lemma A.3.

Step 1 (definition of G). Set

$$b(t) = \min\{a(t), -1/8\}, \qquad q(t) = b(t) - a(t) = -\max\{a(t) + 1/8, 0\},$$

so $|q| \leq 5/16 + 1/8 = 7/16$, $\mathrm{Lip}(q) \leq 1/8$, and $q(t) = 0$ when $a(t) \leq -1/8$. Let $\beta, \chi : [0, \infty) \to [0, 1]$ be

$$\beta(r) = \begin{cases} 1, & r \leq \rho/4, \\ 2 - 4r/\rho, & \rho/4 < r < \rho/2, \\ 0, & r \geq \rho/2, \end{cases} \qquad \chi(r) = \begin{cases} 1, & r \leq \rho/2, \\ 2 - 2r/\rho, & \rho/2 < r < \rho, \\ 0, & r \geq \rho. \end{cases}$$

Let $G(x) = x$ for $x \in B \setminus U$. For $x \in B \cap U$ write $t = t(x)$, $e = e(x)$, $r = \|e\| = \mathrm{dist}(x, \gamma(\mathbb{R}))$, and set

$$\begin{aligned} G(x) &= \gamma(t) + \beta(r) q(t) v(t) + \big((1 - \chi(r))\mathrm{Id} + \chi(r) R_t\big) e \\ &= x + \beta(r) q(t) v(t) + \chi(r)(R_t - \mathrm{Id}) e. \end{aligned} \tag{A.5}$$

Step 2 (G is Lipschitz). Let $x, y \in B$, $\Delta = \|x - y\|$.

(i) $x, y \in U$, $\Delta < \rho$. By (A.1), $|t(x) - t(y)| \leq 16\Delta$; hence $\|e(x) - e(y)\| \leq \Delta + \|\gamma(t(x)) - \gamma(t(y))\| \leq (1 + 16M)\Delta$; and r is 1-Lipschitz. The factors in (A.5) therefore satisfy:

factor	$\beta(r)$	$q(t)$	$v(t)$	$\chi(r)$	$R_t - \mathrm{Id}$	e
bound	1	7/16	1	1	2	ρ
change $\leq$	$4\Delta/\rho$	2Δ	16Δ	$2\Delta/\rho$	$16K_R\Delta$	$(1 + 16M)\Delta$

A product of three factors changes by at most the sum, over the factors, of that factor's change times the bounds of the other two. Hence

$$\|G(x) - G(y)\| \leq \Big[1 + \Big(\frac{7}{4\rho} + 2 + 7\Big) + \big(4 + 16K_R\rho + 2(1 + 16M)\big)\Big]\Delta.$$

(ii) $\Delta \geq \rho$. On B, $\|G(x)\| \leq 1 + 7/16 + 2\rho < 3/2$, so $\|G(x) - G(y)\| \leq 3 \leq (3/\rho)\Delta$.

(iii) $x \in U$, $y \notin U$, $\Delta < \rho$. Both cutoffs satisfy $\beta(r) \leq \chi(r) \leq 2(\rho - r)/\rho$, so

$$\|G(x) - x\| \leq \frac{2(\rho - r)}{\rho}\Big(\frac{7}{16} + 2\rho\Big) = \Big(\frac{7}{8\rho} + 4\Big)(\rho - r).$$

As r is 1-Lipschitz and $r(y) \geq \rho$, $\rho - r \leq \Delta$. With $G(y) = y$: $\|G(x) - G(y)\| \leq (7/(8\rho) + 5)\Delta$.

(iv) $x, y \notin U$. G is the identity.

Step 3 (nonvanishing). We claim

$$\|G(x)\| \geq \rho/4 \qquad (x \in B). \tag{A.6}$$

Outside U: $\|G(x)\| = \|x - \gamma(0)\| \geq \rho$. Inside U:

- $r \leq \rho/2$. Then $\chi = 1$ and $G(x) = (a(t) + \beta(r)q(t))v(t) + R_t e$, with $\langle R_t e, v(t)\rangle = \langle R_t e, R_t u(t)\rangle = \langle e, u(t)\rangle = 0$ and $\|R_t e\| = r$. If $r \leq \rho/4$, then $\beta = 1$ and the coefficient of $v(t)$ is $b(t) \leq -1/8$, so $\|G(x)\| \geq 1/8 > \rho/4$. If $r \geq \rho/4$, then $\|G(x)\| \geq \|R_t e\| \geq \rho/4$.

- $\rho/2 \le r < \rho$. Then $\beta = 0$ and the last term in the first line of (A.5) has norm at most $r < \rho$. If $t > 1$, $\|\gamma(t)\| = a(t) \ge 1/8$, so $\|G(x)\| \ge 1/8 - \rho > \rho/4$. If $t \le 1$, $R_t = \mathrm{Id}$, so $G(x) = a(t)v(t) + e$ with $e \perp u(t) = v(t)$, and $\|G(x)\| \ge r \ge \rho/2$.

Step 4 ($G(x) = x$ when $\|x\| \ge 1/2$). Outside U this is the definition. Inside U,

$$|a(t)| = \|\gamma(t)\| \ge \|x\| - r > 1/2 - \rho > 15/32 > 5/16.$$

Since $a \le 5/16$ everywhere, $a(t) < -5/16$; thus $a(t) \le -1/8$ and $t \le 1$, so $q(t) = 0$ and $R_t = \mathrm{Id}$, and (A.5) gives $G(x) = x$.

Step 5 (the map f). Define $f(x) = -G(x)/\|G(x)\|$. The map $y \mapsto y/\|y\|$ is Lipschitz with constant at most $8/\rho$ on $\{\|y\| \ge \rho/4\}$, so f is Lipschitz by Step 2 and (A.6), and $\|f(x)\| = 1$. If $\|x\| < 1/2$, then $\|f(x) - x\| \ge 1 - \|x\| > 1/2$. Otherwise $G(x) = x$ by Step 4, $f(x) = -x/\|x\|$, and $\|f(x) - x\| = 1 + \|x\| \ge 3/2$. □

Verification checklist

1. **Order of choices.** f, L, δ, D, the intervals I_j and the set S are fixed before A; only n and α depend on A, and one n is used for every color in the proof (Proposition 2.3, Step 0).
2. **Transport identity.** $X_I(g) = X_{I_1}(h_{I,J}g)$ needs admissibility of both g and $h_{I,J}g$ and the exact covariance (2.2) (Step 1; §2.1).
3. **Which pairs are separated.** Types (i)–(iii) in Step 1, and the counts $D(D-1)$ versus D in Step 4; a child against its own parent is only bounded by -1.
4. **Recursion identity** (2.11): admissibility plus (2.1) put $(gT^{(n)})_{I_i}$ in the first case of (2.5) (Step 2); well-foundedness of (2.5) (§2.2).
5. **Cancellation of** α in (2.16) and the arithmetic from Step 5 to (2.8), using (2.4).
6. **Transports lie in** F (Lemma 2.2, Step 4) and fine images are basic (Lemma 2.1, Step 2).
7. **Appendix.** Uniform separation (A.2), especially the case of both parameters tending to $+\infty$ (Lemma A.2, Step 4); nonvanishing (A.6) (Proposition A.1, Step 3); the Lipschitz case analysis (Step 2).
8. **Corollaries.** They rest on companion results not reproduced in the source; see the boxed notes in §3.

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